

SET A

CSE427 Makeup Class Test

Summer 2026

Duration: 20 Minutes

Name:

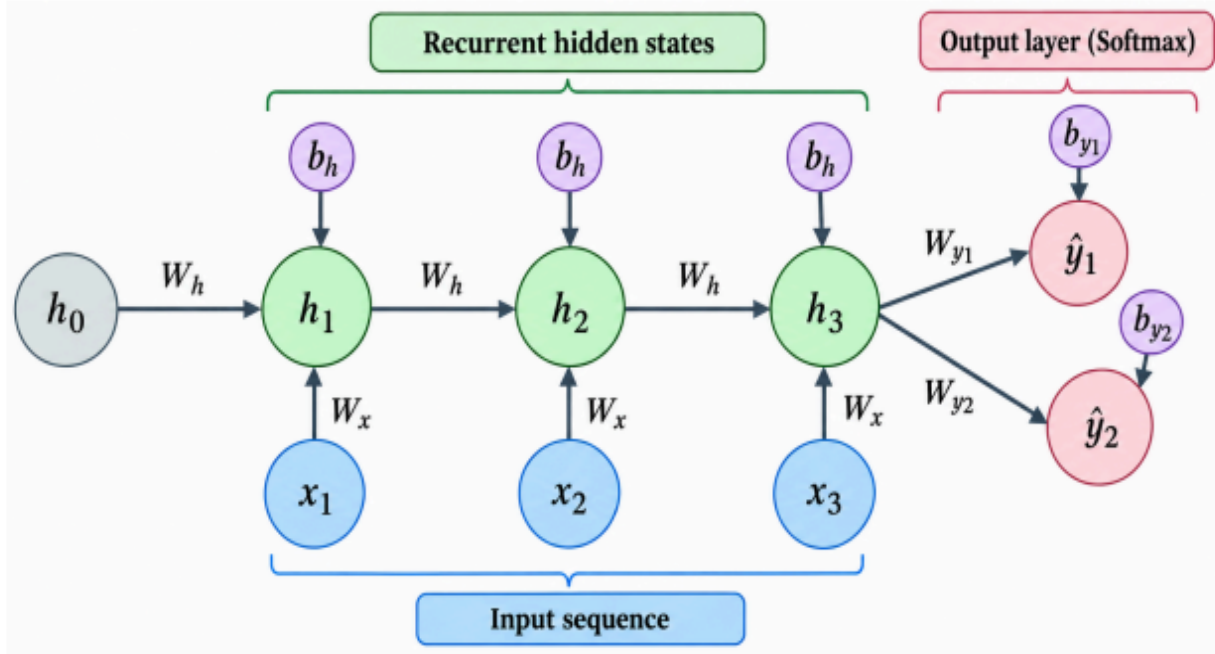
ID:

Sec:

SL:

Question [15 Points]

Consider the following unrolled RNN:



Suppose for the input sequence is $x_1 = 1$, $x_2 = 0$, $x_3 = 1$; the output is $y_1 = 0$, $y_2 = 1$. The hidden state equation for a specific time step (t) is:

$$h_t = \text{sigmoid}(W_x x_t + W_h h_{t-1} + b_h)$$

For the final time step the logit equation is: $z_{(n)} = W_{y(n)} h_3 + b_{y(n)}$

Given Parameters:

h_0 (initial hidden state) = 0

$W_x = 0.5$, $W_h = 0.4$, $b_h = 0.1$

$W_{y1} = 0.6$, $W_{y2} = -0.4$, $b_{y1} = 0.1$, $b_{y2} = -0.1$

For the given input sequence, calculate the prediction and cross-entropy loss.

SET B

CSE427 Makeup Class Test

Summer 2026

Duration: 20 Minutes

Name:

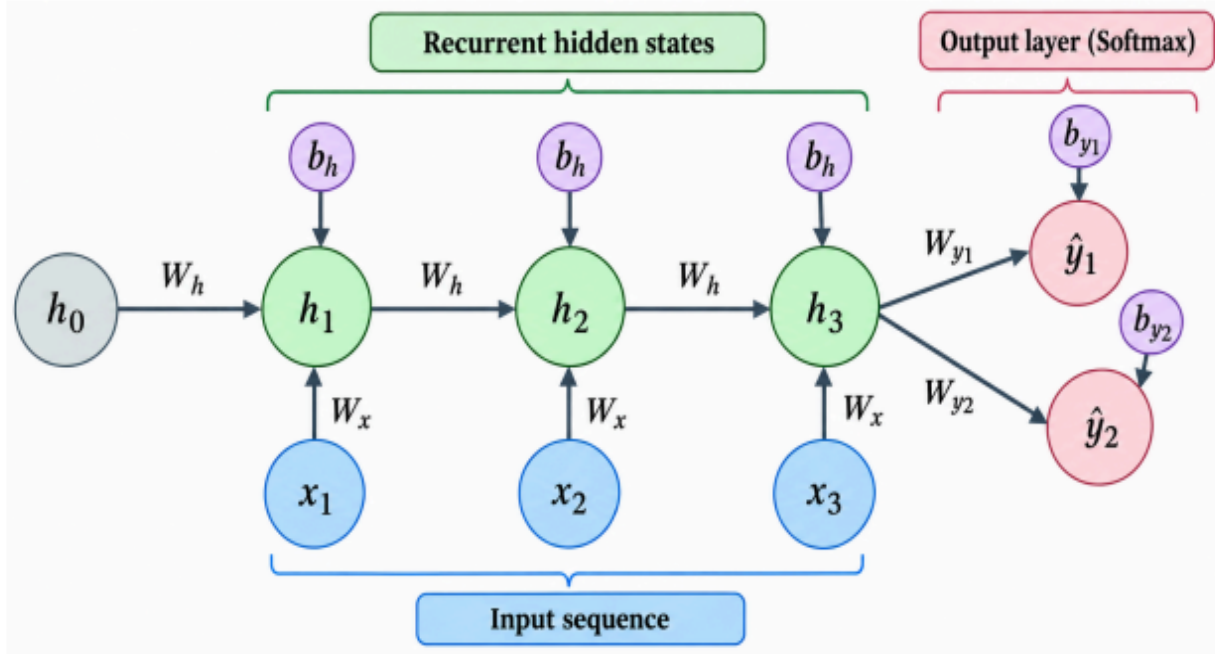
ID:

Sec:

SL:

Question [15 Points]

Consider the following unrolled RNN:



Suppose for the input sequence is $x_1 = 1$, $x_2 = 1$, $x_3 = 0$; the output is $y_1 = 0$, $y_2 = 1$. The hidden state equation for a specific time step (t) is:

$$h_t = \text{sigmoid}(W_x x_t + W_h h_{t-1} + b_h)$$

For the final time step the logit equation is: $z_{(n)} = W_{y(n)} h_3 + b_{y(n)}$

Given Parameters:

h_0 (initial hidden state) = 0

$W_x = 0.5$, $W_h = 0.4$, $b_h = 0.1$

$W_{y1} = 0.6$, $W_{y2} = -0.4$, $b_{y1} = 0.1$, $b_{y2} = -0.1$

For the given input sequence, calculate the prediction and cross-entropy loss.

SET C

CSE427 Makeup Class Test

Summer 2026

Duration: 20 Minutes

Name:

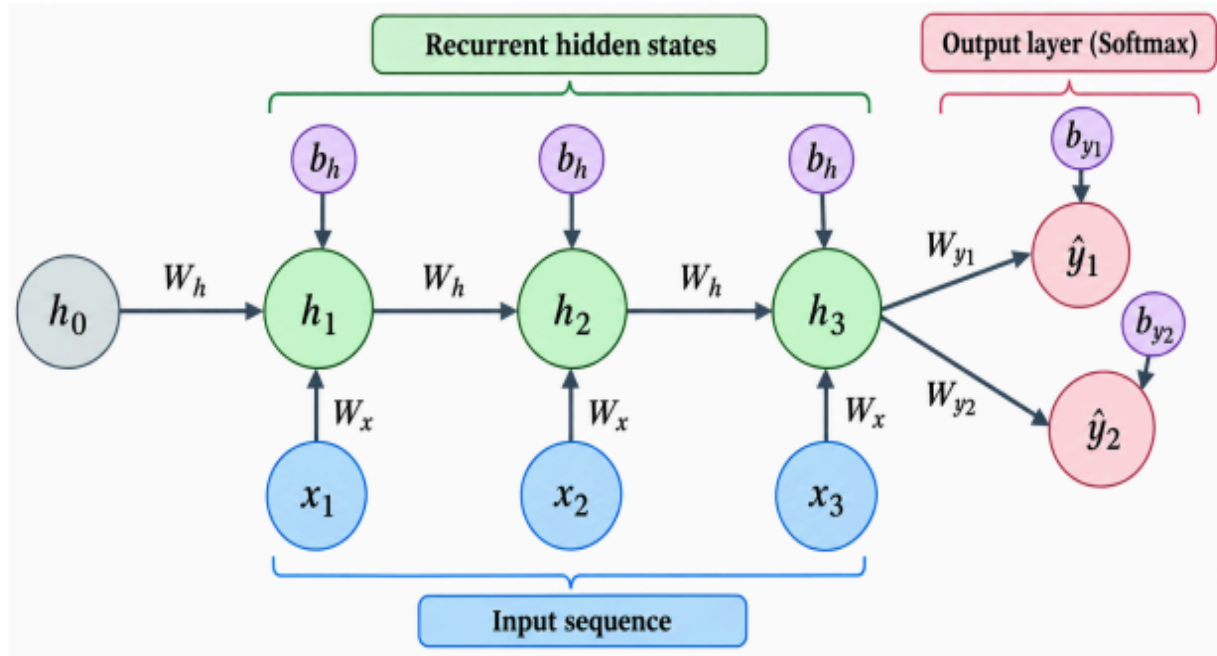
ID:

Sec:

SL:

Question [15 Points]

Consider the following unrolled RNN:



Suppose for the input sequence is $x_1 = 0$, $x_2 = 1$, $x_3 = 1$; the output is $y_1 = 0$, $y_2 = 1$. The hidden state equation for a specific time step (t) is:

$$h_t = \text{sigmoid}(W_x x_t + W_h h_{t-1} + b_h)$$

For the final time step the logit equation is: $z_{(n)} = W_{y(n)} h_3 + b_{y(n)}$

Given Parameters:

h_0 (initial hidden state) = 0

$W_x = 0.5$, $W_h = 0.4$, $b_h = 0.1$

$W_{y1} = 0.6$, $W_{y2} = -0.4$, $b_{y1} = 0.1$, $b_{y2} = -0.1$

For the given input sequence, calculate the prediction and cross-entropy loss.

Solution

Set	Input (x_1, x_2, x_3)	h_1	h_2	h_3	z_1	z_2	$\hat{y}_1$	$\hat{y}_2$	Prediction	Loss _{CE}
A	(1, 0, 1)	$\sigma(0.6)$ = 0.6457	$\sigma(0.3583)$ = 0.5886	$\sigma(0.8354)$ = 0.6975	0.5185	-0.3790	0.7104	0.2896	Class 1	$-\ln(0.2896)$ = 1.2394
B	(1, 1, 0)	$\sigma(0.6)$ = 0.6457	$\sigma(0.8583)$ = 0.7023	$\sigma(0.3809)$ = 0.5941	0.4565	-0.3376	0.6887	0.3113	Class 1	$-\ln(0.3113)$ = 1.1670
C	(0, 1, 1)	$\sigma(0.1)$ = 0.5250	$\sigma(0.81)$ = 0.6921	$\sigma(0.8768)$ = 0.7062	0.5237	-0.3825	0.7122	0.2878	Class 1	$-\ln(0.2878)$ = 1.2455